Narendra M

Backend & Cloud Architect | AI/LLM Engineering

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LinkedIn ↗
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PROFILE SUMMARY

- Backend and cloud architect with 12+ years of experience designing, modernising and scaling distributed platforms across retail supply chain, banking and healthcare.
- Hands-on expertise in Java, Python, Spring Boot, AWS, Kubernetes, Kafka, microservices, event-driven systems and RAG architectures.
- Experienced in prompt engineering and LLM solution integration — translating business needs into effective prompts for OpenAI GPT and Hugging Face models, and partnering with developers to ship generative AI and NLP capabilities into production applications.
- Proven record of improving platform performance, availability, cost efficiency and delivery speed while leading architecture governance, modernisation programmes and cross-team engineering initiatives.
- Strong communicator who explains complex technical concepts to non-technical stakeholders, and collaborates closely with domain experts to turn business requirements into working systems.

TECHNICAL SKILLS

ARCHITECTURE & STRATEGY

Distributed Systems Design · Cloud-Native Architecture · Architecture Reviews & Governance · Modernisation Roadmaps · Platform Extensibility

AI/ML & PROMPT ENGINEERING

Prompt Engineering · Spec-Driven Development (Spec Kit) · AI-Assisted Delivery (Claude, Windsurf) · RAG Architectures · OpenAI GPT Models · LangChain & Hugging Face · NLP Fundamentals · Pandas / NumPy

RELIABILITY & LEADERSHIP

Performance Engineering · High Availability & Failure Isolation · Observability · Mentoring & Design Reviews · Stakeholder Communication

BACKEND & PLATFORMS

Java & Spring Boot · Microservices & REST APIs · Apache Kafka / Event-Driven · Python · Redis / MongoDB

CLOUD & DELIVERY

AWS (EKS, EC2, S3) · Kubernetes & Docker · Terraform & IaC · CI/CD & Automation · GCP

PROFESSIONAL EXPERIENCE

Endava

Oct 2025 – Present

Senior Development Lead · Bangalore, India

- Own end-to-end architecture and technical delivery for healthcare pricing and Medicare calculation platforms on AWS EKS.
- Drive Spec Kit driven (spec-driven) development with Claude as the AI coding agent inside Windsurf, keeping specifications as the source of truth for every feature.
- Define Terraform-based infrastructure and CI/CD patterns adopted across engineering teams.
- Lead architecture governance and cross-team design alignment for HIPAA- and SOC 2-compliant cloud solutions.
- Raise engineering standards for security, resilience and operability across the platform estate.
- Reduced distributed pricing-calculation response time from 45 seconds to under 2 seconds by redesigning processing flows and introducing Redis caching.
- Improved environment provisioning speed by 60% through Terraform-based infrastructure patterns.
- Introduced Spec Kit driven development with Claude and Windsurf, making AI-assisted delivery repeatable and reviewable across the team.
- Increased delivery consistency across teams with standardised CI/CD pipelines.

Tech: Java · Spring Boot · AWS EKS · Redis · Terraform · Kubernetes · CI/CD · Spec Kit · Claude · Windsurf

ANZ

Nov 2023 – Sep 2025

Lead Engineer · Bangalore, India

- Shaped target architecture and execution for wholesale credit platform modernisation across multiple engineering stakeholders.
- Designed Kafka-based event-driven credit workflows to improve scalability, decoupling and operational resilience.
- Designed and built an AI-powered Retrieval-Augmented Generation (RAG) solution in Python for automated document analysis.
- Developed, tested and iteratively refined prompts for OpenAI GPT and Hugging Face-based LLMs, improving output accuracy and relevance.
- Partnered with developers to integrate the RAG/LLM solution into existing enterprise workflows.
- Built a reusable prompt-engineering framework promoting reuse, knowledge sharing and documented best practices across business units.
- Gathered and analysed user feedback on generated outputs to continuously improve prompt effectiveness.
- Mentored engineers and established enterprise observability practices.
- Improved API latency by 35% and reduced cloud spend by 20% through performance tuning, autoscaling and systematic optimisation.
- Delivered the bank's reusable prompt-engineering framework, adopted across multiple business units.
- Tracked advancements in LLM tooling (OpenAI, Hugging Face) to inform platform and tool selection decisions.

Tech: Java · Python · Spring Boot · AWS EKS · Kafka · LangChain · OpenAI GPT · Hugging Face

Target Corporation

Aug 2018 – Nov 2023

Senior Software Engineer · Bangalore, India

- Architected inventory-management systems supporting 10M+ SKUs across 1,800+ stores.
- Designed scalable event-driven services using Java, Spring Boot, Kafka and Kubernetes for high-volume inventory updates across stores and digital channels.
- Led platform migration to AWS and GCP, improving extensibility and operational efficiency.
- Implemented Kubernetes-based deployment and reliability practices for business-critical inventory services.
- Partnered across engineering and business teams to resolve performance and scalability bottlenecks.
- Reduced infrastructure costs by 20% through multi-cloud platform migration.
- Sustained 99.9% platform uptime for business-critical inventory services.
- Scaled inventory processing to 10M+ SKUs across 1,800+ retail stores.

Tech: Java · Spring Boot · Kafka · Kubernetes · AWS · GCP · MongoDB

Philips Innovation Campus

Oct 2016 – Aug 2018

Senior Software Engineer · Bangalore, India

- Developed secure healthcare-monitoring services and APIs using Java, Spring, Spring Security and JWT.
- Designed Docker- and Kubernetes-based platforms for connected healthcare devices.
- Improved deployment consistency and scalability across connected-device services.
- Delivered secure, standards-compliant APIs for connected healthcare monitoring.
- Introduced containerised deployment practices for medical-device platforms.

Tech: Java · Spring · Spring Security · JWT · Docker · Kubernetes

Cognizant Technology Solutions

Dec 2013 – Sep 2016

Programmer Analyst · India

- Delivered CVS/Caremark pharmacy solutions that improved refill adherence and patient engagement.
- Enhanced prescription-management systems with notification and alert capabilities for critical customer workflows.
- Improved prescription refill adherence through targeted notification workflows.

Tech: Java · J2EE · SQL · Web Services

PROJECT EXPERIENCE

Healthcare Pricing & Medicare Calculation Platform

Oct 2025 – Present

Endava

A cloud-native pricing engine running on AWS EKS that performs distributed Medicare and commercial pricing calculations for business-critical healthcare workflows, built to HIPAA and SOC 2 standards. Delivered using Spec Kit driven (spec-driven) development with Claude as the AI coding agent inside Windsurf, so every feature traces back to an executable, reviewed specification.

Impact: Cut pricing-calculation response time from 45 seconds to under 2 seconds and improved environment provisioning speed by 60%. Spec-driven development with Claude and Windsurf shortened the path from requirement to reviewed implementation and kept specs, tests and code in sync.

Tech: Java · Spring Boot · AWS EKS · Redis · Terraform · Kubernetes · Spec Kit · Spec-Driven Development · Claude · Windsurf · CI/CD

AI Document Analysis — RAG & Prompt Engineering Platform

2024 – Sep 2025

ANZ

A Python-based Retrieval-Augmented Generation solution that automates analysis of complex credit and legal documents, integrated directly into existing enterprise banking workflows.

Impact: Automated manual document review across multiple use cases and established the reusable prompt-engineering framework adopted across business units.

Tech: Python · LangChain · OpenAI GPT · Hugging Face · Vector Search · REST APIs

Wholesale Credit Platform Modernisation

Nov 2023 – Sep 2025

ANZ

Modernisation of legacy wholesale credit systems into Spring Boot microservices on AWS EKS, with Kafka-based event-driven credit workflows replacing tightly coupled batch processes.

Impact: Improved API latency by 35% and reduced cloud spend by 20% while increasing operational resilience of critical credit processes.

Tech: Java · Spring Boot · AWS EKS · Kafka · Kubernetes · Observability

Enterprise Inventory Management Platform

Aug 2018 – Nov 2023

Target Corporation

Event-driven inventory platform tracking 10M+ SKUs across 1,800+ retail stores and digital channels, migrated to a multi-cloud AWS and GCP footprint.

Impact: Sustained 99.9% uptime for business-critical inventory services while reducing infrastructure costs by 20%.

Tech: Java · Spring Boot · Kafka · Kubernetes · AWS · GCP · MongoDB

Connected Healthcare Monitoring Services

Oct 2016 – Aug 2018

Philips Innovation Campus

Secure backend services and APIs for connected healthcare monitoring devices, deployed on containerised Docker and Kubernetes platforms.

Impact: Delivered secure, scalable services underpinning connected medical-device monitoring.

Tech: Java · Spring · Spring Security · JWT · Docker · Kubernetes

EDUCATION

Bachelor of Engineering

Computer Science & Engineering

Jawaharlal Nehru National College of Engineering, Shimoga · Visvesvaraya Technological University · 2013 ·

AWARDS & RECOGNITION

AWS Certified Developer — Associate

Amazon Web Services ·

Certification covering cloud-native application development on AWS.

Spring Professional Certification

VMware / Pivotal ·

Certification covering Spring Framework and Spring Boot application development.

PERSONAL DETAILS

DATE OF BIRTH	NATIONALITY	LOCATION
	Indian	Bangalore, Karnataka
LANGUAGES	INTERESTS	
English, Kannada, Hindi	Distributed Systems, Generative AI, Cloud Cost Engineering	

References available on request.